

Seismology

Pupil sheet

Name: _____

Date: _____

1. What I see on the screen

The screen shows five cards and an **event counter**. The **Magnitude** card is $|a| = \sqrt{a_x^2 + a_y^2 + a_z^2}$ — **and it includes gravity**, so at rest it reads about 1 g, not zero.

The counter, though, subtracts gravity first and counts a shake when the departure from rest passes 0.2 g. A shake is counted **once**, however long it lasts.

The counter is not exported: I write it on paper, before and after each interval.

2. What I do on my own

First I predict. Five vertical palms will raise the counter by about _____; five sideways pushes, just as strong, by about _____. One long two-second shake, by about _____.

because _____

Test 1 — direction. Five knocks of the same strength in each direction, with silence between the sets.

Direction	Before	After	Difference	Seconds
Vertical				
Horizontal				

Test 2 — duration. Same direction, same strength. Once one long shake, once five short knocks.

Kind	Before	After	Difference	Seconds
One shake, 2 s				
Five short knocks				

I divide the difference by the seconds: _____ : _____ = _____ per second. What number should come out if the threshold was exceeded the whole time?

Test 3 — two intervals. Two different places, or two different times of day. **The same duration for both,** or the comparison means nothing.

Place / time	Before	After	Difference	Minutes

3. The conclusion I write

I predicted that, and the data show that _____

Direction did not matter because _____

The counter does not count earthquakes, but _____

Exit ticket

The counter reached 34. What did it count — and what did it not?
